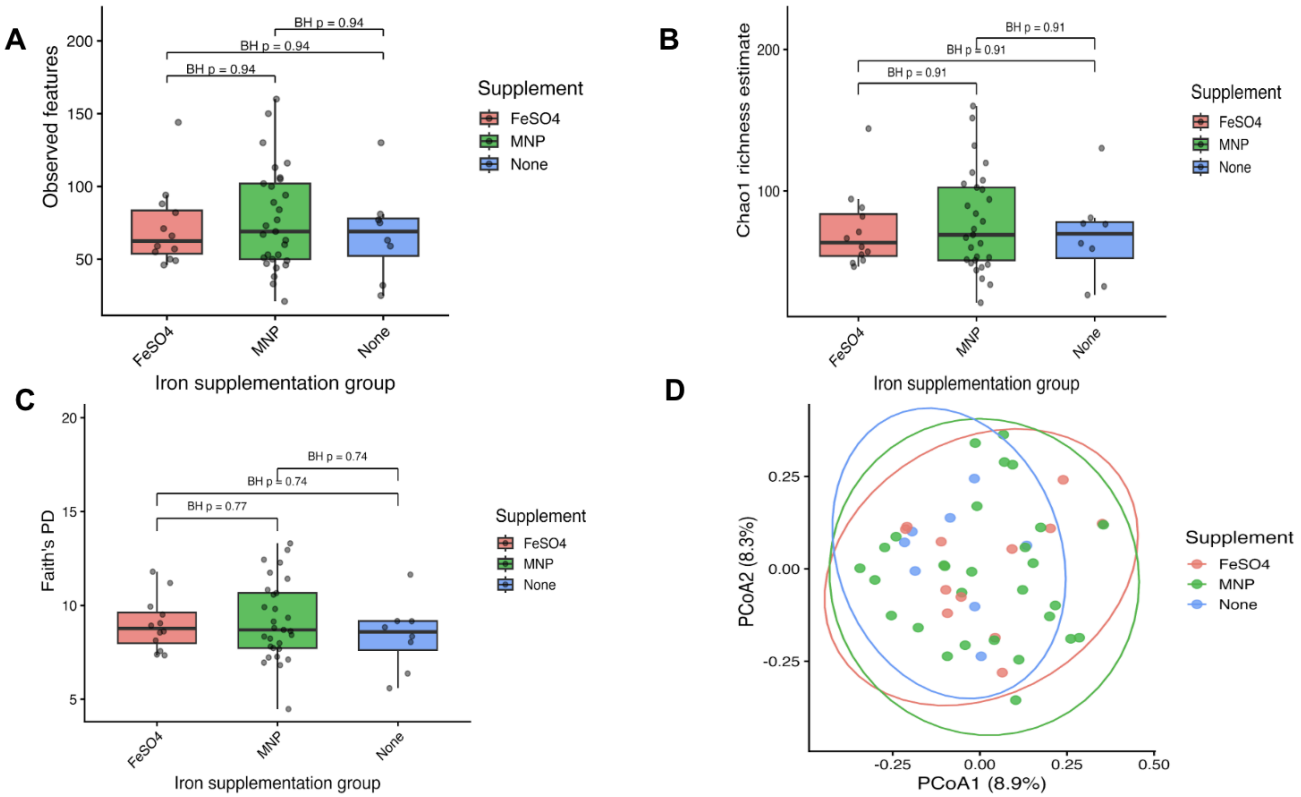
Supplementary Material

Genus	s.FeSO4	s.MNP	s.None	Index	Stat	P-Value
Fusobacterium	1	1	1	7	0.5822225	N/A
Lactobacillus	1	1	1	7	1	N/A
Akkermansia	1	1	0	4	0.6491705	0.255
Campylobacter	1	1	0	4	0.4081425	0.61
Escherichia-Shigella	1	1	1	7	1	N/A
Bifidobacterium	1	1	1	7	1	N/A
Bacteroides	1	1	1	7	0.9914892	N/A

Supplementary Figure 1. Indicator Species Analysis Results. Analysis was done using the multipatt function with the supplement groups being set as predictors (31). Taxa with ≤ 0.1 abundance were filtered out.



Supplementary Figure 2. Alpha and beta diversity across iron supplementation groups. (A–C) Alpha diversity metrics comparing gut microbiome diversity across $FeSO_4$, micronutrient powder (MNP), and no supplementation groups, including (A) observed features, (B) Chao1 richness, and (C) Faith's phylogenetic diversity (22, 23). Statistical significance was assessed using the Kruskal-Wallis test, with Benjamini-Hochberg correction applied for p-values. Boxplots represent the interquartile range with median values, and individual points correspond to individual samples. (D) Beta diversity analysis based on Bray–Curtis dissimilarity visualized using principal coordinates analysis (PCoA) (24). Each point represents a sample, colored by the supplementation group, with ellipses indicating 95% confidence interval (27).